

RAJAT RAVINDRA KOORSE

AI-ML Engineer

Bangalore, India | +91 8618908564 | rajatkourse.rk@gmail.com | [linkedin.com/in/rajatkourse](https://www.linkedin.com/in/rajatkourse)

PROFESSIONAL PROFILE

- Over 2.5+ years of professional experience in developing scalable, high-performance AI-ML applications with a focus on GenAI and multimodal perception.
- Expertise in architecting LLM-driven pipelines, RAG-based architectures, and Agentic workflows to enhance development velocity and system intelligence.
- Skilled in engineering deep learning-based perception modules for multi-modal sensor fusion (Camera, LiDAR, Radar) and automated validation platforms.
- Proven ability to deliver production-grade AI solutions using Python, Flask, and AWS, consistently reducing model inference latency through optimization and quantization.
- Experience in designing synthetic data generation workflows and automated testing suites to improve model robustness and validation coverage.
- Strong foundation in full-stack development, REST API integration, and performance tuning for real-time inference environments.

Technical Skills

- Programming Languages:** Python, Java
- Database:** MySQL, MongoDB
- Tools:** VS Code, GitHub, RStudio, Tableau, AWS, GCP, Docker, Flask, FastAPI (basic)
- Machine Learning:** Algorithms, Deep Learning, Neural Networks, Data Analysis, Computer Vision, NLP, Semantic Search, Synthetic Data Generation, Prompt Engineering, RAG, LLM APIs, Multimodal AI, Agent-style Workflows, PyTorch, TensorFlow, LangChain

Professional Experience

TATA ELXSIAI-ML Engineer | Mar 2024 - Present

Project 1: GenAI-Powered ADAS Scenario Intelligence Platform

- Led the end-to-end development of a GenAI platform to automate critical ADAS scenario retrieval and scene understanding pipelines for autonomous validation.
- Architected LLM-driven multimodal processing engines capable of digesting visual, textual, and tabular sensor data to extract behavioral patterns and edge cases.
- Designed and deployed 20+ automated testing workflows using Agentic AI, accelerating validation timelines for Software-Defined Vehicle (SDV) modules by 40%.
- Engineered production-ready microservices using Python and Flask to interface seamlessly with commercial LLM and vision APIs under high-concurrency constraints.
- Conducted extensive model performance tuning and quantization techniques, driving down inference latency by 30% while retaining model evaluation accuracy.
- Containerized application workflows using Docker and established scalable deployment patterns on AWS ECS, ensuring 99.9% pipeline availability.
- Prepared detailed technical design documentation and established system performance baselines to ensure alignment with enterprise-grade deployment standards.

Project 2: Intelligent Perception & Sensor Fusion Framework

- Engineered deep learning perception networks tailored for processing multi-modal sensor data streams (Camera, LiDAR, and Radar) to elevate environment mapping fidelity.
- Constructed specialized synthetic data generation pipelines to dynamically generate rare target profiles and edge-case hazards, boosting model accuracy on unseen data by 25%.
- Integrated a highly responsive RAG (Retrieval-Augmented Generation) infrastructure over enterprise vector databases, allowing internal teams to query dense sensor logs using natural language.
- Developed and exposed customized REST APIs to handle high-throughput telemetry streams, enabling smooth integration with backend validation systems.
- Implemented robust continuous validation logic to track model performance drift across recurring regression cycles, achieving an aggregate 95% bug-resolution rate prior to staging.
- Designed real-time metrics dashboards within internal full-stack platforms to surface active model evaluation stats, inference timings, and edge-case highlights.

TecCera Innovations – AI-ML Trainer | Jul 2023 – Mar 2024

- Designed and delivered comprehensive, hands-on technical training curriculum covering core Artificial Intelligence and Machine Learning paradigms utilizing Python, TensorFlow, and Keras.
- Led deep-dive instructional sessions on classical machine learning algorithms, deep neural network architectures, computer vision basics, and natural language processing techniques.
- Guided over 10+ end-to-end, real-world development projects, navigating teams through the entire engineering pipeline from exploratory data analysis and algorithm selection to production-grade model deployment.
- Mentored software developers and engineering students on scalable AI architecture choices, data preprocessing strategies, hyperparameter tuning, and model evaluation metrics.
- Established industry best practices for writing clean, modular ML code, ensuring proper model versioning, reproducibility, and optimal resource utilization during training phases.
- Evaluated student code and architectural designs, delivering constructive technical feedback to align academic projects with enterprise-level code quality and engineering standards.

Freelance Full Stack Developer | Jul 2020 – Mar 2024

- Delivered 15+ full stack and backend-driven projects for clients.
- Built scalable backend services, data pipelines, and API integrations.

Projects

- **Emotion-Based Book Recommendation System:** Developed a full stack system using AI and machine learning to detect emotions and personalize book recommendations.
- **MCE Staff Loan Management System:** Created a web application with AI-ML techniques in Java, PHP, and SQL to automate loan processing workflows.
- **Virtual Health Assistant:** Collaborated on an application that offers personalized health insights using natural language processing and machine learning for real-time health advice and preventive care.

Education

MTech in Artificial Intelligence and Machine Learning

- BITS Pilani (2025–Present)

Bachelor of Engineering in Information Science and Engineering

- Malnad College of Engineering (2019–2023)

Certifications

- "Python for Data Science" course offered by NPTEL
- "Cybersecurity Fundamentals" program on LinkedIn Learning
- "Data Streaming Engineer Certification" by Confluent
- "Advanced Project Management Certificate" from Learntube.ai

Achievements

- Extra Mile Award (Tata Elxsi Ltd): Honored for relentless commitment, driving exceptional project delivery, and optimizing AI validation workflows within the Transportation Business Unit.
- Best Team of the Quarter (Tata Elxsi Ltd): Recognized alongside the ADAS Core Engineering group for successfully delivering the GenAI-driven scenario intelligence platform ahead of key client timelines.
- Technical Mentor Distinction (TecCera Innovations): Awarded the 'Best Instructor' title for achieving a 95% milestone success rate across 10+ student-led machine learning deployment projects.